Fiscal Year:	FY 2009	**Task Last Updated:**	FY 05/03/2010
PI Name:	Barcellos-Hoff, Mary Helen Ph.D.		
Project Title:	NSCOR: Mechanisms of HZE Damage and Repair in Human Epithelial Cells		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
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Space Biology Special Category:	None		
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Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
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No. of Post Docs:	6	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: NCE to 5/29/2009 has been approved, per A. Chu/ARC (10/08) NOTE: NCE to 5/31/2009, from 11/30/2008, is in process, per A. Chu/ARC (9/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kronenberg, Amy (Lawrence Berkeley National Laboratory) Parvin, Bahram (Lawrence Berkeley National Laboratory) Chatterjee, Aloke (Lawrence Berkeley National Laboratory) Rydberg, Bjorn (Lawrence Berkeley National Laboratory) Yaswen, Paul (Lawrence Berkeley National Laboratory) Costes, Syvlain (Lawrence Berkeley National Laboratory)		
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Performance Goal Text:	
Task Description:	NASA's Biomedical Critical Path Roadmap defines the carcinogenic risks of radiation exposure as one of only four Type I risks identified. Estimates of cancer risk from human travel in space should be based on a mechanistic understanding of complex effects that different types of radiation exposure elicit. Because most adult human solid tumors are epithelial in origin and because radiation effects are dictated by cellular genotype and phenotype, studies of HZE particle radiation biology using a physiologically relevant model will be the most informative regarding risk. We have developed methods for culturing human mammary epithelial cells (HMEC) derived from normal tissue as both monolayer and multicellular aggregates, which enable study of phenotypes and pathways that may be uniquely human, epithelial or multicellular. We have studied how controlled cellular and microenvironment conditions affect early responses to HZE particle radiation-induced DNA damage in terms of repair, foci formation and gene expression, and persistent radiation-induced phenotypes and pathways that precede neoplasia. The potential for inducing genomic instability, which may contribute to its carcinogenic effect, has been measured in both finite lifespan HMEC . Together, these surrogate endpoints will enable functional evaluation of neoplastic potential as a function of radiation quality, energy and dose. These data will be integrated at two levels: by theoretical modeling of the physical events leading to DNA damage and by systems biology modeling of critical pathways. The intent of this 5-year project NSCOR at Lawrence Berkeley National Laboratory Life Sciences Division is to provide a comprehensive examination of HZE effects from the initial damage, to early cellular HMEC responses, to persistent functional precursors of carcinogenesis. We have addressed the mechanisms, nature, and frequency of DNA damaging events; mechanisms of DNA repair and misrepair, early signal transduction mechanisms; immediate and long-term, reversible and irreversible gene expression changes; cellular remodeling and reorganization; potential mechanisms of tissue "repair" and matrix effects; and the cellular and molecular mechanisms of HZE/proton induced progression to a neoplastic phenotype.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Current modeling of risk relies on comparing the relatively biological effectiveness of low LET radiation to high LET radiation effects, thus providing an earth benefit in the description and understanding of low dose reference radiation effects. In the context of the NSCOR, we have developed image analysis programs to quantify DNA damage response (e.g. radiation-induced foci) and specific organelles (i.e. centrosomes) using high-throughput imaging bioinformatics. The characterization of non-malignant human mammary epithelial cells responses to ionizing radiation provides mechanistic insights to the processes relevant to human health risks from diagnostic and therapeutic radiation exposures. The NSCOR has supported the development of a new media that enables human mammary epithelial cells to be maintained longer in culture.
	Accomplishments Rationale: Epithelial cells depend on signals from the microenvironment to establish the requisite polarity for functional differentiation. Because most adult human solid tumors are epithelial in origin and because radiation effects are dictated by cellular genotype and phenotype, studies of HZE particle radiation biology using a physiologically relevant model will be the most informative regarding risk. The LBNL NSCOR developed methods for culturing human mammary epithelial cells (HMEC) derived from normal tissue as both monolayer and multicellular aggregates to study phenotypes and pathways that may be uniquely human, epithelial or multicellular. We examined how controlled cellular and microenvironment conditions affect early responses to HZE particle radiation-induced DNA damage in terms of repair, foci formation and gene expression, genomic instability, and persistent radiation-induced phenotypes and pathways that precede neoplasia. These surrogate endpoints were used to evaluate the neoplastic potential as a function of radiation quality, energy and dose. Aim 1. We hypothesized that HZE particles such as 1GeV/u Fe-ions induce more persistent DNA damage than sparsely ionizing gamma rays (Rydberg et al., 2005). Studies using micronuclei supporting this hypothesis were published in Radiation Research (Groesser et al., 2007). To further test this idea, gamma-H2AX and 53BP1 radiation-induced foci (RIF) were measured in proliferating HMEC after exposure to low doses of sparsely and densely ionizing radiation. Preliminary results suggest that radiation induced gamma-H2AX and 53BP1 RIF in HMEC have different resolution dynamics as a function of radiation quality. While RIF numbers after gamma-ray irradiation returned to control level after about 12h, Fe-ion induced foci were still present at 24h. However, persistent RIF following HZE particles were eventually removed 24 – 48h after irradiation under the conditions used. The slower disappearance of foci after Fe-ion exposure could be due to the fact that HZE particles induce more complex DNA damage in the core of the ion track, which will be tested in future experiments using different ions and LET. We speculated that the persistent gamma-H2AX foci in Fe-ion irradiated non-proliferating cells could be due to the fact that not all DSB repair pathways for rejoining complex DSBs are available to the cell in G0/G1-phase or that damaged cells in a cycling population could be eliminated more efficiently via an apoptotic pathway. Apoptosis measurements in proliferating and non-proliferating MCF10A cells after exposure to different radiation qualities indicate that the apoptotic rate is very low in these cells and removal of damaged cells by apoptosis can only have a minor effect on our repair kinetic data. Another possibility is that the remaining gamma-H2AX foci represent chromatin remodeling rather than DSBs. These studies being prepared for publication (Groesser et al. in preparation). Since energy is randomly deposited along high-LET particle paths, RIF along these paths should also be randomly distributed. A novel imaging approach shows that RIF were located preferentially at the interface between high and low DNA density regions, and were more frequent than predicted in regions with lower DNA density. The same preferential nuclear location was also measured for RIF induced by 1 Gy of low-LET radiation. This deviation from random behavior was evident only 5 min after irradiation for phosphorylated ATM RIF, while gamma H2AX and 53BP1 RIF showed more pronounced deviations up to 30 min after exposure. These data are published in PLoS Computational Biology (Costes et al., 2007). Following this high content analysis on fixed specimen showing that radiation induced foci (RIF) predominantly form in the euchromatin and at the interface between euchromatin/heterochromatin, we are using live cell imaging in cell line stably transfected with 53BP1-GFP from collaboration with Dr. Asaithamby

Task Progress:

Aroumougame from UT Southwestern NSCOR. The first movies obtained during NSRL07C and NSRL08A indicate two different types of foci movements. Many foci appear to move in a diffusive manner (Brownian motion), where as others show active movement with velocities that can reach 2 microns/min (Costes et al. in preparation).

We published one of the first comparisons of foci from gamma vs HZE irradiated cells (Costes et al., 2006). We have measured imaging properties of foci on and off tracks (e.g. foci frequency, foci size, foci intensity, and spatial organization of foci). Presumably, foci outside tracks are generated by delta-rays, where as foci along tracks are generated by the high-LET component of HZE. The conclusion of this analysis is a clear difference in foci formation kinetic for these two components. As previously reported, the maximum number of low-LET foci form over the course of 30 min to 1 hour. In contrast, foci along tracks appear within 2 min post-IR with a maximum frequency of approximately 0.7 foci/micron at these early times. Initial foci size and intensity are similar for both types of foci but low-LET foci have a slight size increase reaching a maximum 10 min post-radiation, high-LET foci seem to have a three-fold size increase over the first 30 min post-IR (Costes et al. in preparation).

Aim 2. While to date, many studies of ionizing radiation have focused on its effects on primary DNA structure, comparatively few have focused on its effects on gene methylation or phenotype. Such epigenetic phenomena have been increasingly implicated in carcinogenesis in human cells and tissues, and in some cases, may represent predominant mechanisms of oncogenesis. HMEC cultures spontaneously yield at low frequencies (1x10-5 to 1x10-8) variant p16(-) cell populations that are capable of long term growth (50-100 total population doublings), and which are susceptible to genomic instability associated with telomere dysfunction. We determined that 2 Gy X-irradiation causes increases in the appearance of p16(-) cells with long term growth potential. Flow cytometry confirmed that the differences in BrdU incorporation and senescence associated beta-galactosidase expression in the treated and untreated populations were statistically significant. Both sparsely and densely ionizing irradiation increased the frequency and rate with which HMEC escaped p16-associated stasis and acquired long-term growth potential inindependent experiments with HMEC from four individuals. We are currently preparing these data for publication and exploring the mechanistic underpinnings of this phenomenon (Mukhopadhyay et al. in preparation).

Our studies reporting that disrupted morphogenesis is a result of irradiated cells undergoing transforming growth factor ß 1 (TGFß) mediated epithelial to mesenchymal transition (EMT) were published in Cancer Research (Andarawewa et al., 2007). TGFß has many functions that orchestrate the response to DNA damage (Andarawewa et al., 2007). We compared the EMT induced by eqi-toxic doses of 1Gy of 1 GeV/amu 56Fe to 2Gy of 137Cs gamma radiation in both monolayer and 3D. Both exposures resulted in EMT in response to TGFß in monolayer and disrupted 3D morphogenesis when embedded in Matrigel. Radiation dose responses show that this effect has a very low threshold in that a single exposure of 3-100 cGy gamma radiation elicits the ‘same’ phenotypic switch, consistent with a non-targeted effect. TGF-ß induced EMT in irradiated HMEC is mediated by MAPK pathway, which crosstalks with integrins via reactive oxygen species (ROS). To test whether ROS could play a role in the EMT phenotype, we measured cellular ROS levels in irradiated, TGFß treated cells. We observed that double treated HMEC (184v and MCF10A) have sustained generation of ROS and increased mitochondria compared to single treatments (Andarawewa et al. in preparation).

Aim 3. We showed that exposure of finite lifespan HMEC in growth arrested monolayer culture to x-rays or Fe ions elicited severe karyotypic instability, defined as at least 3 subpopulations of cells per colony, in the clonal descendants of ~20% of clones. This phenotype was not seen in any of the unirradiated colonies. Although severe karyotypic instability was seen in colonies arising after exposure to the lowest dose of Fe ions (0.5 Gy), there was no evidence of a dose response and it was not evident after exposure to <2 Gy of x-rays. These data are now published in Radiation Research (Sudo et al., 2008). A dose response was seen for centrosome aberrations (CA) defined as 3 or more centrosomes/cell for the progeny of cells exposed in monolayers to x-rays or Fe ions. Unirradiated clones had a baseline frequency of 0.92% CA. Although there was no evidence for an RBE>1 for the induction of CA following irradiation in monolayer, irradiation of the same finite lifespan HMEC as growth arrested acinar structures in Matrigel increased colony forming ability of the descendents of the irradiated cells following exposure to x-rays, but eliminated the induction of centrosome aberrations. In contrast, exposure of these HMEC as preformed acinar structures provided no protection against Fe ion-induced cell killing or against the induction of centrosome aberrations suggesting that there may be an increased RBE for each of these endpoints in cells exposed to Fe ions in a more tissue-like architecture.

We have now published in Cancer Research that CA actually precede radiation-induced genomic instability in HMEC (Maxwell et al., 2008). Although CA induction is a linear dose response from 10-200 cGy, evidence of genomic instability in clonal assays showed a threshold between 10 and 50 cGy, suggesting another level of control on their persistence. We demonstrate that TGFß is a key mediator of genomic surveillance in human epithelial cells. TGFß did not prevent CA but rather deletes genomically unstable cells by p53 dependent apoptosis. Thus the targeted effect on centrosomes is countered by non-targeted signaling that limits the burden of potentially neoplastic cells. This model is now being extended to HZE exposures to test whether the differences seen in 3D as a function of radiation quality is based on induction efficiency or deletion efficiency.

Aim 4. We postulated that carcinogenesis can be thought as an emerging phenomenon arising from the perturbation of an organ’s homeostasis (Barcellos-Hoff and Costes, 2006). Emergent phenomena typically result from the interactions of individual entities (Barcellos-Hoff and Costes, 2006). By definition, a system with emerging properties cannot be reduced to its individual entities and as such “the whole is greater than the sum of its parts”. To address this concept, we began developing agent-based models (ABM), which are computer programs that represent a system as a collection of autonomous decision-making entities called agents. Each agent individually assesses its situation and makes decisions on the basis of a set of contextual rules. In the case of representing cells within an organism, agents may execute various behaviors appropriate for the system such as proliferation, differentiation or death. Such models have proven to be very useful in predicting emerging properties from complex systems. Using an agent-based model, we could predict growth population of post-stasis HMEC by simply assuming a 26 hour doubling time and their response to radiation as a function of clonogenic death that accelerates the emergence of variants. New studies using such approach are underway with very promising results. These studies complement those underway in the Tufts NSCOR and will provide a means of integrating biological data into a meaningful format for risk modeling.

In support of the overarching use of cell biology tools in radiation quality studies, we developed the BioSig imaging bioinformatics framework (Parvin et al., 2005). Information can be obtained from BioSig through either ad hoc queries or query by example. A previously developed 3D segmentation method quantitative analysis for 3D cell culture models (Raman et al., 2007) has been improved for its accuracy through multi-class labeling and geometric constraints (Parvin et al., 2007). The end result is that foci can be quantified in full 3D from volumetric representation. Once the data is analyzed, rendered surfaces are generated, and stored in the database. The analysis package has been designed to export

	the same feature-based representation in terms of morphology, structure, and localization. The schema supports more than 400 inter-related computed representations on a cell-by-cell basis. One recent novelty has been integration of computational and informatics content directly within the database (https://). The net result has been direct visualization of user-selectable features through the BioSig imaging bioinformatics system. As a result, multi-dimensional representations can be visualized through heatmaps for hypothesis generation and direct visualization (Han et al., 2008). Summary The NSCOR data demonstrate that radiation quality affects types and persistence of DNA damage, genomic instability, cell fate, the microenvironment and responses to soluble signals. Potential interactions among these effects likely determine the probability of progressing to cancer ((Barcellos-Hoff, 2007). The LBNL NSCOR studies have shown that in general high vs low LET radiation is more effective in eliciting many biological processes like GIN and EMT that promote cancer development. However the fact that such responses do not exhibit typical dose dependence prevents the use of classic RBE comparisons. We propose that these studies motivate further examination of whether individual targeted and non-targeted phenomena (e.g. damage response, phenotype, genomic instability) interact in a unique manner to drive neoplastic transformation following high LET exposure.
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